MILITARY SPECIFICATION

COMPRESSOR, AIR, POWER DRIVEN, PORTABLE 5 CFM, 175 PSI

1. SCOPE

1.1 Scope. This specification covers a power driven, 5 cfm, 175 psi portable air compressor unit.

1.2 Classification. The compressor units shall be of the following types:

Type I - Electric motor driven.
Type II - Gasoline engine driven.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-3600	Compressors, Air and Gas (except Oxygen), Packaging of

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U.S. Military Property

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

National Electrical Code No 70

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch St, Boston, MA 02110.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Code for Unfired Pressure Vessels - Section VIII
Power Test Code for Displacement Compressors, Vacuum Pumps and Blowers - PTC-9

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th St, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA MG-1	Standard for Motors
NEMA IC-1	Standard for Industrial Controls

(Application for copies should be addressed to National Electrical Manufacturers' Association, 155 East 44th St, New York, NY 10017.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE J607	Small Stock, Two Stroke and Four Stroke Cycle, Gasoline Engine Test Code
SAE J608	Minimum Identification Markings for Small, Air-Cooled Engines

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., Two Pennsylvania Plaza, New York, NY 10001.)

DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Occupational Safety and Health Act (OSHA) Standards with Supplemental Revisions

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

3. REQUIREMENTS

3.1 Description. The compressor unit shall be 2-wheel mounted and shall consist essentially of a two-stage, air-cooled, reciprocating compressor or an air-oil cooled, multi-stage rotary compressor, either type to be equipped with power unit, controls, air receiver, air hose and accessories described herein.

3.2 Preproduction model. The contractor shall furnish a compressor unit for examination and test before starting production. Examination and test shall be those specified herein. The preproduction model shall be identical with the units the contractor proposes to furnish in fulfillment of the contract.

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3.3 Materials. Materials shall be as specified herein. Material not definitely specified shall be of a type and quality used for the intended application in commercial practice.

3.4 Operating temperature. The compressor unit shall be capable of performing as specified herein in ambient temperatures of minus 25° F to plus 120° F.

3.5 Safety. Guards shall be provided to protect the operator from all exposed moving parts. High temperature parts and electrically energized parts shall be guarded, insulated, or so located that contact by the operator will not occur during normal operation. Sharp edges and hazardous projections shall be avoided. Equipment must be consistent with applicable "OSHA" standards.

3.6 Receiver. The air receiver shall be designed, constructed and stamped in accordance with ASME Code for Unfired Pressure Vessels for a working pressure of 200 psi. The receiver shall be equipped with an automatic resetting pressure relief valve set to relieve at 190 psi ±10 psi and a drain valve located in the lowest point of the receiver. The air outlet shall be located at or above the receiver centerline and shall be supplied with an air regulator adjustable from 0-175 psi. Gages shall be supplied to indicate receiver pressure and regulated outlet pressure. The receiver shall have a minimum capacity of 20 gallons.

3.7 Air hose. The compressor unit shall be supplied with a 25 foot, 5/16 I.D., oil-resistant air hose with external NPT end fittings capable of operating at a pressure of 175 psi. One end of the hose shall be adapted to connect directly to the outlet side of the regulator.

3.8 Mounting. The compressor unit shall be mounted on 2 semi-pneumatic (zero pressure) tire and wheel assemblies. A support leg shall be provided to hold the compressor unit in a level position. A handle shall be provided for one-man moving.

3.9 Power unit.

3.9.1 Type I, electric motor driven. The compressor shall be driven by a 115/230 volt, 60 hertz, single-phase, 1½ HP (minimum) dripproof electric motor conforming to NEMA Standard MG-1. The motor shall be a continuous duty type and shall have sufficient starting torque to start the compressor under all conditions specified herein. Unless otherwise specified (see 6.2) the motor shall be connected and labeled to operate on 115 volts.

3.9.1.1 Electric power cable. A 25 foot, 3-conductor, type SO cable shall be supplied. The third conductor shall serve as an equipment ground and the compressor shall be capable of operation without the third conductor connected.

3.9.1.2 Starter. A manual motor starter conforming to NEMA Standard IC-1, installed in a NEMA 1 enclosure shall be provided. Wiring shall be in accordance with the NFPA National Electrical Code.

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3.9.1.3 Pressure regulation. A pressure actuated electrical stop-start control shall be supplied. The control shall be adjusted to stop the motor when the pressure reaches 175 psi, plus 0 psi, minus 10 psi and to start the motor when the pressure drops to 140 ±10 psi.

3.9.2 Type II, gasoline engine driven. The compressor shall be driven by a 4 HP (minimum) air-cooled, 4-cycle gasoline engine capable of producing a net continuous horsepower rating at the applicable speed as required by the compressor and conforming to SAE J607 and SAE J608. The engine shall be provided with a fuel tank of sufficient capacity to insure a minimum of one hour continuous running at rated load operation and a manual cranking system. All engine accessories provided commercially as standard equipment shall be furnished.

3.9.2.1 Pressure regulation. A constant-speed type pressure regulator shall be supplied that is designed to load the compressor when the receiver pressure drops to 140 ±10 psi and to unload the compressor when the receiver pressure reaches 175 psi plus 0 psi, minus 10 psi.

3.10 Drive. The compressor shall be either direct drive or multiple V-belt drive. When a direct drive is used, a flexible coupling shall be installed between the power unit and the compressor. When a multiple V-belt drive is used, means shall be provided for adjusting the V-belts.

3.11 Compressor. The compressor shall deliver 5 cfm of free air at a pressure of 175 psi as determined by the ASME Power Test Code for Displacement Compressors, Vacuum Pumps, and Blowers, and shall not exceed the manufacturer's recommended continuous operating speed. The compressor shall be balanced so that vibration will be minimized. The compressor shall be air-cooled or air-oil-cooled and shall be designed for a minimum life of 1,000 hours. The compressor shall be provided with a system to automatically unload the compressor when shut down.

3.11.1 Air filter. The compressor shall be provided with a dry-type air filter with sufficient capacity for 8 hours operation without exceeding an air restriction which could adversely affect the rated output of the compressor.

3.11.2 Intercooler and aftercooler. The compressor shall be provided with an intercooler and aftercooler as required in order to maintain the receiver outlet air temperature at ambient plus 100° F or less.

3.11.3 Air-oil separator. Rotary compressors shall be supplied with an air-oil separating system.

3.12 Tilted operation. The compressor unit shall be designed to deliver full capacity at rated pressure when operating in any tilted plane up to 15° from the horizontal.

3.13 Lubrication. The compressor unit shall be designed to operate efficiently when lubricated with standard military lubricants (see 6.3).

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3.14 Treatment, painting and marking. The portions of the compressor unit normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard practice and the finish shall be gray, color No 16473, according to FED-STD-595. The compressor unit shall be marked for identification according to MIL-STD-130.

3.15 Shipping data plate. A corrosion-resistant metal plate containing the length, width, height, curb weight, gross vehicle weight and the center of balance of the unit shall be provided in an accessible location.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Inspection classification. Inspections shall be classified as follows:

a. Preproduction inspection (see 4.2.1).

b. Quality conformance inspection (see 4.2.2).

4.2.1 Preproduction inspection. The preproduction model shall be examined in accordance with para 4.3 and prepared and tested in accordance with para 4.4, 4.4.1, 4.4.2, 4.4.3, and 4.4.5.

4.2.2 Quality conformance inspection. Each production model shall be examined in accordance with para 4.3 and prepared and tested in accordance with 4.4 and 4.4.4.

4.3 Examination. The compressor shall be examined during and after construction in accordance with the manufacturer's established inspection system and as may otherwise be required to verify compliance with the non-operational requirements of this specification. The examination shall verify that all components, operating controls, safety devices, accessories and gages are supplied and properly installed and that the item conforms to the applicable requirements of this specification. The compressor unit manufacturer shall have the responsibility for verifying that the receiver has been constructed, tested, and inspected in accordance with the ASME Code. ASME Code stamp or permanently attached labels on the receiver shall be accepted as evidence of compliance.

4.4 Preparation for test. Prior to beginning any test specified herein, the compressor unit shall be completely serviced with standard military lubricants.

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4.4.1 Performance test. The compressor shall be tested to determine the capacities specified in 3.11 in accordance with ASME PTC-9. The duration of this test shall be sufficient to record all data required by PTC-9, but in no case shall the test duration be less than 1 hour. Inability of the compressor to meet the performance requirements or to operate satisfactorily shall be cause for rejection.

4.4.2 Tilted operation test. Following the successful completion of the performance test, the compressor shall be operated for a period of 30 minutes in each of the 4 tilted positions specified in 3.12. Inability of the compressor to meet the performance requirements or to operate satisfactorily in any of the tilted positions shall be cause for rejection.

4.4.3 Functional test. The compressor shall be operated for a period of 80 hours in increments of not less than 8 hours, delivering not less than rated pressure and capacity. When a gasoline engine driven compressor is specified, an auxiliary fuel supply may be used or shut-down periods of not more than 10 minutes will be allowed for refueling. Normal lubrication maintenance shall be performed during downtime but there shall be no parts replaced. During 8 hours of this test, the receiver discharge will be adjusted to cause the compressor to operate through a complete loading-unloading or stop-start cycle 10 times per hour. The capacity, pressure, compressor speed and temperature of air entering the receiver shall be recorded during each hour of operation. Failure of the compressor to meet the performance requirements specified in 3.11 and 3.11.2 or malfunction or failure of any part or controls shall be cause for rejection.

4.4.4 Operational test. Each compressor, excluding the preproduction model, shall be tested for a period of not less than 30 minutes to verify satisfactory operation of the compressor. During this test all necessary adjustments in controls shall be made as required to insure that performance requirements relating to compressor output, regulation and speed will be met.

4.4.5 Mobility. After the completion of the functional test, move the compressor for a distance of not less than 100 yards, turning in complete circles to the right and left and in a straight line for at least 50 yards. Move the compressor over a 6 inch drop onto a concrete surface not less than three times. After completing the mobility test, repeat the performance test in 4.4.1. Inability of the compressor to be moved by one man, inability to be carted in a straight path, inability to be turned to the right or left, or fracture or permanent deformation of any component, or inability to perform as specified in 3.11 shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packing and packaging. The compressor unit shall be preserved, packed and packaged in accordance with MIL-C-3600 (see 6.2).

6. NOTES

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6.1 Intended use. This compressor is intended for use in filling stations, vehicle maintenance shops, hobby shops and in any non-hazardous location where compressed air is required.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type required (see 1.2).

c. When electric motors shall be connected other than as specified (see 3.9.1).

d. Level of preservation, packing and packaging (see 5.1).

6.3 Lubricants. When requested by the contractor, the contracting officer should make available a list of lubricants applicable to the equipment covered in this specification.

6.4 Definition. The term compressor unit is defined as the complete air compressor package including power unit, compressor, fans, electrical equipment, receiver, controls, mobility equipment and accessories described in this specification.

Custodian:
Air Force - 84

Preparing activity:
Air Force - 84

Project Number:
4310-F059

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

MIL-C- 83980 (USAF)

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE. AF-WP-O-MAY 64 505

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